```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

Struct Node {
   Int id;
   Char name[50];
   Float parking, steering, rules, total;
   Struct Node *left, *right;
};

// ===== INSERT STUDENT MODULE =====
Struct Node* insert(struct Node* root, int id, char name[], float p, float s, float r) {
   If (!root) {
      Struct Node* n = (struct Node*)malloc(sizeof(struct Node));
      n->id = id;
      strcpy(n->name, name);
      n->parking = p;
      n->steering = s;
      n->rules = r;
      n->total = (p + s + r) / 3;
      n->left = n->right = NULL;
      return n;
   }
```

```
   If (id < root->id)
      Root->left = insert(root->left, id, name, p, s, r);
   Else if (id > root->id)
      Root->right = insert(root->right, id, name, p, s, r);

   Return root;
}

// ===== DISPLAY STUDENTS MODULE =====
Void display(struct Node* root) {
   If (root) {
      Display(root->left);
      Printf("\nID: %d\nName: %s\nParking: %.1f\nSteering: %.1f\nRules: %.1f\nTotal:
%.1f\n",
            Root->id, root->name, root->parking, root->steering, root->rules, root->total);
      Printf("----------------------------\n");
      Display(root->right);
   }
}

// ===== SEARCH STUDENT MODULE =====
Struct Node* search(struct Node* root, int id) {
   If (!root || root->id == id)
      Return root;

   Return (id < root->id) ? search(root->left, id) : search(root->right, id);
```

```
}

// ===== DELETE STUDENT MODULE =====
Struct Node* deleteNode(struct Node* root, int id) {
  If (!root)
    Return root;

  If (id < root->id)
    Root->left = deleteNode(root->left, id);
  Else if (id > root->id)
    Root->right = deleteNode(root->right, id);
  Else {
    If (!root->left) {
      Struct Node* temp = root->right;
      Free(root);
      Return temp;
    }

    If (!root->right) {
      Struct Node* temp = root->left;
      Free(root);
      Return temp;
    }

    Struct Node* t = root->right;
    While (t->left)
```

```
        T = t->left;

      Root->id = t->id;
      Strcpy(root->name, t->name);
      Root->parking = t->parking;
      Root->steering = t->steering;
      Root->rules = t->rules;
      Root->total = t->total;

      Root->right = deleteNode(root->right, t->id);
   }

   Return root;
}

// ===== WEAK SKILL MODULE =====
Void weakSkill(struct Node* s) {
   If (!s)
      Return;

   Printf(“Weak Skill: “);

   If (s->parking <= s->steering && s->parking <= s->rules)
      Printf(“Parking\n”);
   Else if (s->steering <= s->parking && s->steering <= s->rules)
      Printf(“Steering\n”);
```

```
    Else
        Printf(“Rules\n”);
}

// ===== LICENSE ELIGIBILITY MODULE =====
Void license(struct Node* s) {
    If (!s)
        Return;

    If (s->total >= 60 && s->parking >= 50 &&
        s->steering >= 50 && s->rules >= 50)
        printf(“License Status: Eligible\n”);
    else
        printf(“License Status: Not Eligible\n”);
}

// ===== MAIN MODULE =====
Int main() {
    Struct Node* root = NULL;
    Int ch, id;
    Char name[50];
    Float p, s, r;

    While (1) {
        Printf(“\n========== Driving Skill System ==========\n”);
        Printf(“1. Add Student Record\n”);
```

```
Printf(“2. Display All Students\n”);
Printf(“3. Search Student by ID\n”);
Printf(“4. Delete Student Record\n”);
Printf(“5. Analyze Student Performance\n”);
Printf(“6. Exit\n”);
Printf(“Enter your choice (1-6): “);
Scanf(“%d”, &ch);

Switch (ch) {
  Case 1:
    Printf(“\n--- Enter Student Details ---\n”);
    Printf(“Enter Student ID: “);
    Scanf(“%d”, &id);

    Printf(“Enter Student Name: “);
    Scanf(“ %[^\n]”, name);

    Printf(“Enter Parking Score (0-100): “);
    Scanf(“%f”, &p);

    Printf(“Enter Steering Score (0-100): “);
    Scanf(“%f”, &s);

    Printf(“Enter Traffic Rules Score (0-100): “);
    Scanf(“%f”, &r);
```

```
    Root = insert(root, id, name, p, s, r);
    Printf("Student added successfully!\n");
    Break;

Case 2:
    Printf("\n--- Student Records ---\n");
    Display(root);
    Break;

Case 3: {
    Printf("Enter Student ID to search: ");
    Scanf("%d", &id);

    Struct Node* x = search(root, id);

    If (x) {
        Printf("\nStudent Found:\nName: %s\nTotal: %.1f\n",
            x->name, x->total);
    } else {
        Printf("Student not found.\n");
    }
    Break;
}

Case 4:
    Printf("Enter Student ID to delete: ");
```

```
    Scanf("%d", &id);

    Root = deleteNode(root, id);
    Printf("Record deleted (if ID existed).\n");
    Break;

Case 5: {
    Printf("Enter Student ID for analysis: ");
    Scanf("%d", &id);

    Struct Node* x = search(root, id);

    If (x) {
        Printf("\n--- Performance Analysis ---\n");
        weakSkill(x);
        license(x);
    } else {
        Printf("Student not found.\n");
    }
    Break;
}

Case 6:
    Printf("Exiting program...\n");
    Exit(0);
```

```
        Default:
            Printf("Invalid choice! Please try again.\n");
      }
   }
}
```